PREPARED FOR

Nxtra by Airtel

A Bharti Airtel Group Company · India's Largest Data Center Network

RE POWER ASSESSMENT

THE REWARD FRAMEWORK

Renewable Energy Assessment & Deployment

"Transforming your largest liability — power — into a long-term competitive advantage"

PREPARED BY

Refex Energy

REWARD Framework v2.6

₹8,420 Cr
10-YR SAVINGS
Live Portfolio

3.8 Yrs
PAYBACK
₹4,880 Cr CAPEX

1.69 MT
CO_2 AVOIDED
Per year

2029
RE100 TARGET
2 Yrs Ahead

February 2026 · Confidential

The Problem: The Data Center Grid Trap

Why inaction is the most expensive strategy

Cost Volatility

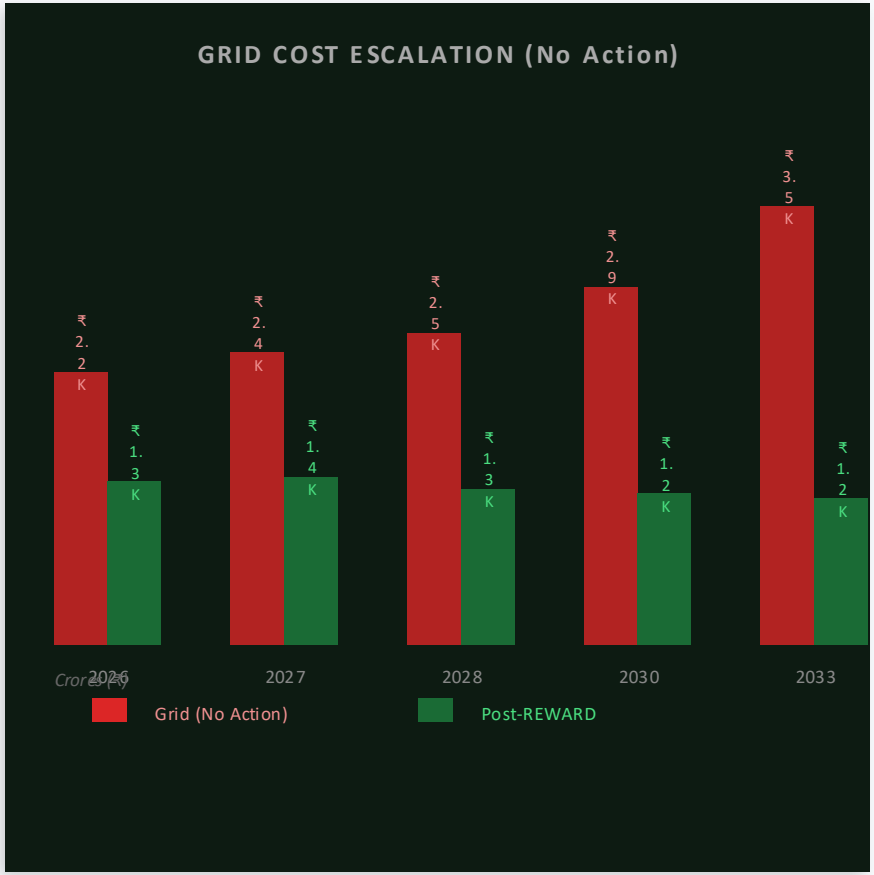
Grid tariffs rising 7% p.a. — ₹7.50/kWh today → ₹13.81/kWh by 2035. ₹2,207 Cr/yr at immediate risk.

Efficiency Ceiling

PUE optimisation has diminishing returns. Average PUE of 1.57 offers headroom but standard cooling is tapped out.

ESG Pressure

Hyperscaler clients demanding 100% RE-powered colocation in RFPs. 82% gap to RE100 commitment by 2031.



THE REWARD FRAMEWORK

Six Phases from Grid Dependency to Renewable-Primary Power

R — AUDIT

Refex Initial Engagement

72-hr baseline audit

Phase 1

E — DESIGN

Energy Renewable Profile

Solar·Wind·H_2 mix

Phase 2

W — MODEL

Wide-Scale Load Mapping

BESS & dispatch sync

Phase 3

A — FINANCE

Assessment & Economics

LCOE + 10yr alpha

Phase 4

R — PROCURE

Resource Optimisation

Procurement & PPA

Phase 5

D — DEPLOY

Deployment & Savings

Zero-downtime go-live

Phase 6

Phase R — Refex Initial Engagement: Baseline Audit

72-hour load data audit · PUE profiling · Grid cost mapping · Carbon baseline

R | E | W | A | R | D

₹2,207 Cr
ANNUAL GRID SPEND
At ₹7.50/kWh avg HT tariff

2,943 GWh
ANNUAL CONSUMPTION
Live portfolio at PUE 1.57

1.57 avg
PORTFOLIO PUE
Range: 1.54 – 1.60 across campuses

18%
CURRENT RE MIX
82% gap to RE100 commitment

CAMPUS-BY-CAMPUS BASELINE

Campus	IT Load	PUE	kWh/yr	Grid Cost
Pune I	96 MW	[illegible]	2,943 GWh	₹362 Cr
Mumbai I	55 MW	1.55	748 GWh	₹561 Cr
Chennai II	40 MW	1.54	540 GWh	₹405 Cr
Manesar I	22 MW	1.55	299 GWh	₹224 Cr
Bangalore	20 MW	[illegible]	281 GWh	₹211 Cr
Kolkata	10 MW	[illegible]	140 GWh	₹105 Cr

KEY FINDINGS

- Peak tariff hours (18:00–23:00) account for 38% of total spend — ₹838 Cr/yr at ₹9.20/kWh
- PUE headroom: 1.57 → 1.40 target would save 36 MW equivalent = ₹376 Cr/yr
- Carbon baseline: 2.41 MT CO_2/yr at 0.82 kg/kWh grid emission factor
- Full 660 MW pipeline at risk: ₹6,937 Cr/yr if RE100 strategy not locked now

Phases E & W — Energy Profiling & Wide-Scale Load Mapping

Custom RE mix · BESS dispatch · 24/7 grid displacement strategy

SOLAR

75 %

LCOE ₹2.45/kWh

- 320 MWp capacity (South + West hubs)
- GHI 5.4 kWh/m²/day Chennai/Pune/Mumbai
- PLF 21–24% · TOPCon 700Wp bifacial
- IT load (68%) — direct solar feed daytime

WIND

15 %

LCOE ₹3.20/kWh

- 80 MW (North + East: Manesar, Kolkata)
- Hub wind 6.8 m/s · IEC Class IIA 3.3MW
- CU 28–32% · Night dispatch via BESS
- Cooling load (22%) — BESS + wind covers

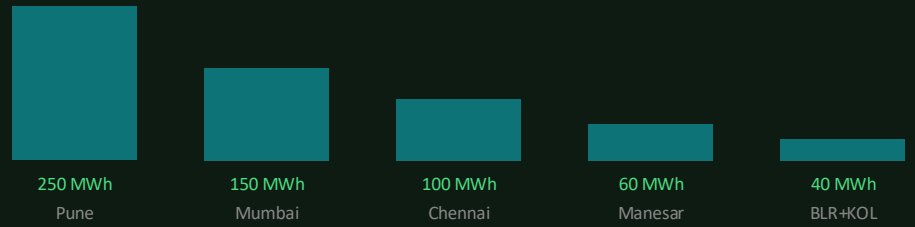
HYDROGEN FC

10 %

LCOE On-site H_2

- 40 MW capacity · India's 1st DC deployment
- H_2-ready fuel cell — 18:00–06:00 dispatch
- Overhead + peak backup (10% of load)
- 99.991% uptime vs. Tier III 99.982% req.

BESS INTEGRATION — 600 MWh TOTAL (LFP · 93.5% RTE · 15-yr life)



70%
Grid Displacement

2,060 GWh
RE Generation/yr

99.991%
Uptime Achieved

₹838 Cr
Peak Cost Avoided

Phase A — Assessment & Economic Modelling

LCOE vs. grid escalation · 10-Year Green Alpha · Regulatory benefits · CAPEX payback

CAPEX INVESTMENT

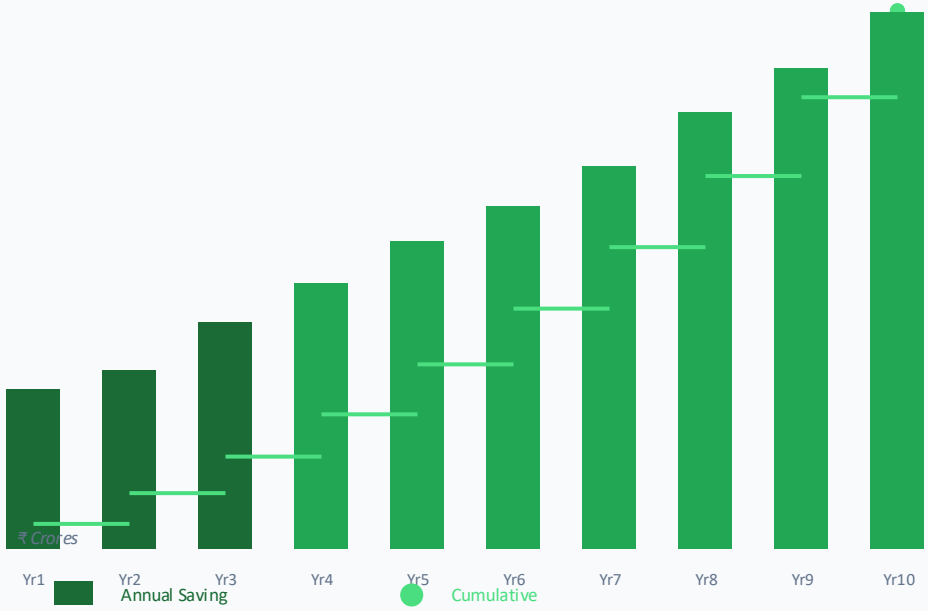
Solar Plants (320 MWp)	₹1,920 Cr
BESS (600 MWh LFP)	₹1,800 Cr
Wind Farms (80 MW)	₹560 Cr
Hydrogen FC	₹420 Cr
Grid Integration	₹180 Cr
TOTAL	**₹4,880 Cr**

18% procurement saving via Refex = –₹878 Cr vs. open market

YEAR 1 NET SAVINGS BREAKDOWN

Grid energy displaced (70%)	+₹728 Cr
PUE improvement 1.57→1.40	+₹376 Cr
Peak demand avoided + REC	+₹291 Cr
Less: O&M (all RE systems)	–₹285 Cr
NET YEAR 1 SAVING	**₹888 Cr**

10-YEAR GREEN ALPHA PROJECTION



₹18,486 Cr
10-YEAR GREEN ALPHA

3.8 Yr Payback
≤₹4,880 Cr CAPEX

₹888 Cr Year 1
Net OpEx Saving

Phase D — Campus Deployment Roadmap

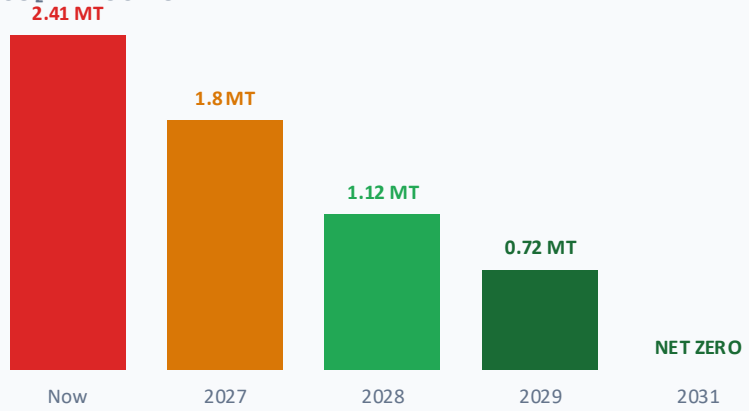
Sequenced zero-downtime rollout · Pune first (highest ROI) → Full portfolio by Q3 2027

	Q2 2026	Q3 2026	Q4 2026	Q1 2027	Q2 2027	Q3 2027
Pune I (96 MW)		230 MWp Solar + 250 MWh BESS	₹362 Cr/yr 2.8yr			
Mumbai I (55 MW)			90 MWp Solar + 150 MWh BESS	₹216 Cr/yr 3.2yr		
Chennai II (40 MW)				60 MWp Solar + H_2-FC + 100 MWh	₹162 Cr/yr 3.5yr	
Manesar I (22 MW)					20 MWp Solar + 20 MW Wind + 60 MWh	₹85 Cr/yr 4.0yr
	₹362 Cr/yr Q3 2026 Pune live	₹578 Cr/yr Q4 2026 + Mumbai	₹740 Cr/yr Q1 2027 + Chennai	₹825 Cr/yr Q2 2027 + Manesar	₹888 Cr/yr Q3 2027 Full live portfolio	₹63 Cr 4.2yr
HYDERABAD (200 MW)	⚡ HYDERABAD (200 MW planned): RE-First design — embed 500 MWp captive solar into civil blueprints NOW to save ₹400–600 Cr in retrofit and unlock <₹2.40/kWh LCOE					

Sustainability Impact & Compliance Status

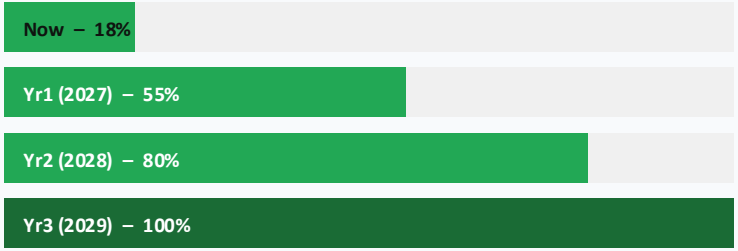
Net Zero 2031 · RE100 by 2029 · 1.69 MT CO_2/yr avoided · Diesel-free by 2028

CO_2 REDUCTION PATHWAY



CO_2 Scope 2 (MT/yr)

RE100 PROGRESS



RE100 achieved by 2029 — 2 years ahead of public commitment

COMPLIANCE & CERTIFICATION STATUS

ISO 50001 Energy	ACTIVE	ISO 14001 Environment	ACTIVE
RE100 Membership	ACTIVE	Net Zero 2031	ON TRACK
TIA 942 Tier III/IV	CERTIFIED	MEITY Certified	CERTIFIED
H_2-FC (India 1st)	DEPLOYED	Open Access (4 States)	IN PROGRESS
ISO 27001 Security	CERTIFIED	PCI DSS	CERTIFIED
SOC 1&2 Type II	CERTIFIED	RPO Compliance	PARTIAL

🏆 CIO Choice Award — Colocation Category · 6 Consecutive Years

Priority Recommendations & Next Steps

Four actions to unlock ₹8,420 Crores in 10-year savings

01 IMMEDIATE

Execute Pune Campus PPA — Q2 2026

230 MWp Solar + 250 MWh BESS. ₹362 Cr/yr from Day 1. 2.8yr payback. Lock ₹2.45/kWh before FY27 tariff revision.

02 HIGH

Hyderabad 200 MW — Design RE-First Now

Embed 500 MWp captive solar in civil blueprints before ground-break. Saves ₹400–600 Cr in retrofit. <₹2.40/kWh LCOE.

03 STRATEGIC

Scale H_2-FC Across All 14 Campuses by 2028

Eliminate ₹85 Cr/yr diesel DG costs. 100% NOx-free. Strengthen RE100 credentials in hyperscaler RFPs.

04 FINANCIAL

Green Finance via Airtel Group Balance Sheet

Green bonds + SLL at 150–200 bps below commercial rates. Saves ₹490–650 Cr CAPEX. Payback 3.8yr → 3.2yr.

REWARD INITIATIVE · SUMMARY SCORECARD

Financial Impact	Sustainability Impact	Technical Performance
10-Year Green Alpha **₹8,420 Cr**	CO_2 Avoided (live) **1.69 MT/yr**	Grid Displacement **70% Year 1**
Full 660 MW Pipeline **₹26,400 Cr**	CO_2 Avoided (pipeline) **4.2 MT/yr**	RE Capacity Installed **440 MW**
Year 1 Net Saving **₹888 Cr**	10-Year Cumulative CO_2 **21.5 MT**	BESS Storage **600 MWh LFP**
CAPEX Total **₹4,880 Cr**	RE100 Achievement **2029 (2yr early)**	Hydrogen FC **40 MW (scalable)**
Payback Period **3.8 Years**	Net Zero Target **2031 On Track**	System Uptime **99.991%**
RE LCOE vs. Grid **₹2.68 vs ₹7.50**	Diesel DG Eliminated **100%**	First Campus Live **Q3 2026 (Pune)**
Peak Cost Avoided **₹838 Cr/yr**	PUE Improvement **1.57 → 1.40**	Full Portfolio Live **Q3 2027**
Procurement Saving **₹878 Cr (18%)**	Equivalent Trees **7.7M per year**	PPA Duration **25 Years Fixed**

Ready to begin? → Contact Refex Energy to schedule your 72-hour REWARD Audit · Refex Energy · REWARD@refexenergy.com